



OFFLINE-FIRST SECURITY ENGINEERING

VNAGY Use Case — Financial Behavioral Anomaly Detection

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1. Executive Summary

This document describes how the **VNAGY** Framework conceptually supports financial environments through behavior-driven analysis of processes, signals, and heuristics. **VNAGY** operates offline-first, without integration into core banking systems and without access to sensitive financial data. The goal is to reduce operational, security, and regulatory risk by detecting early anomalies in system behavior.

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2. Business Context

Financial institutions rely on predictable, stable, and auditable system behavior. Behavioral anomalies often precede incidents such as fraud, unauthorized access, system compromise, or operational disruption.

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3. Problem Statement

Typical behavioral risks in financial systems include:

- Unusual API usage patterns
- Memory-only processes interacting with critical modules
- Irregular timing profiles
- Network bursts to unknown destinations
- Abnormal CPU/memory behavior
- Unexpected process lineage

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4. VNAGY Role

VNAGY:

- Analyzes process and signal behavior
- Correlates heuristics (CPU, network, timing, lineage, memory)
- Detects deviations from normal operational patterns
- Assigns deterministic risk levels
- Operates offline-first without integration

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5. Workflow Overview

- A financial system generates an event.
- VNAGY collects the signal via the local event bus.
- Heuristics evaluate behavioral deviation.
- VNAGY assigns a risk level.
- The system receives a recommendation.

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6. Expected Outcomes

- Early anomaly detection
- Reduced operational risk
- Improved auditability
- Increased transparency
- Enhanced stability

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7. Risk Reduction

VNAGY reduces:

- Operational risk
- Security risk
- Regulatory risk
- Reputational risk

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8. Conclusion

VNAGY provides stable, predictable, and secure behavior-driven analysis for financial environments without integration or access to sensitive data.

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9. Licensing Reference

Subject to **VNAGY-PSDL-1.3** licensing constraints.

[https://github.com/VNAGY-sec-ViktorijaN/VNAGY-Framework/blob/main/VNAGY/LICENSE/VNAGY%20Minimal%20License%20\(PSDL%E2%80%911.3\).pdf](https://github.com/VNAGY-sec-ViktorijaN/VNAGY-Framework/blob/main/VNAGY/LICENSE/VNAGY%20Minimal%20License%20(PSDL%E2%80%911.3).pdf)

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